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Paleomagnetic study of Sicily: consequences for the deformation of Italian and African margins over the last 100 million years

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Paleomagnetic analysis of Cenozoic sediments from the Iblean platform (12 sites, 100 samples) has provided four new poles. These data are combined with those available from volcanic series to yield a new apparent polar wander path for "stable" Sicily since the Upper Cretaceous. This curve, when compared to a revised curve for Africa (using a selection of available data points), demonstrates a 15° rotation of Sicily with respect to Africa during the Plio-Pleistocene. This rotation accounts for the extensional tectonic regime that prevails in the Pelagian Sea and which is evidenced by "en relais" rifts (Malta, Pantelleria, Linosa), associated with alkaline volcanism of Pliocene and Quaternary age. Paleomagnetic directions from various areas of the Italian Peninsula also indicate anticlockwise rotations with diverse amplitudes. The chronology of these rotations is consistent with a deformation of the western Italian margin during progressive suturation from north to south of the Corsican-Sardinian block since the end of the Oligocene.

1. Introduction

Paleogeographic reconstructions show that Sicily is an emerged fragment of the North African margin. Between Southern Italy and Tunisia, Sicily represents the eastern termination of the Maghrebide chain and the southern termination of the Apennines. Paleogeographical units such as a characteristic succession of basins and margins have been sometimes correlated with similar structures in southern peninsular Italy [1,2]. The complex tectonic environment (presence to the north of a volcanic arc, tangential tectonics until the Late Miocene and Pliocene, extensional tectonics to the south and to the east during the Plio-Quaternary) has led us to carry out a paleomagnetic study on Tertiary sediments from the Ragusa Iblei platform ("stable" Sicily), in order to detect relative movements with respect to Africa.

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2. Sicily in its geological context

Sicily is separated from the African domain by the Pelagian Sea (Fig. 1), which is underlain by 20 km of continental crust. This domain shows evidence of extensional tectonics characterized by the existence of "en relais" rifts affecting the Upper Miocene to Quaternary series (Pantelleria, Linosa, Malta, Tunisian grabens) with a roughly N120 direction, and often associated with an active alkaline volcanism. The depth of most of these grabens reaches frequently more than 1500 m, and the sedimentation during the Pliocene compensates for the faults' dip-slip, showing a continuous extension phenomenon. The "cumulated" thickness of the Plio-Quaternary sediments ranges from 1000 to 3000 m. The subsidence rate of the Tunisian grabens is greater or almost equal, but their bathymetric gap is less pronounced due to the greater quantity of terrigenous sedimentation. These grabens are linked one to another by trans-

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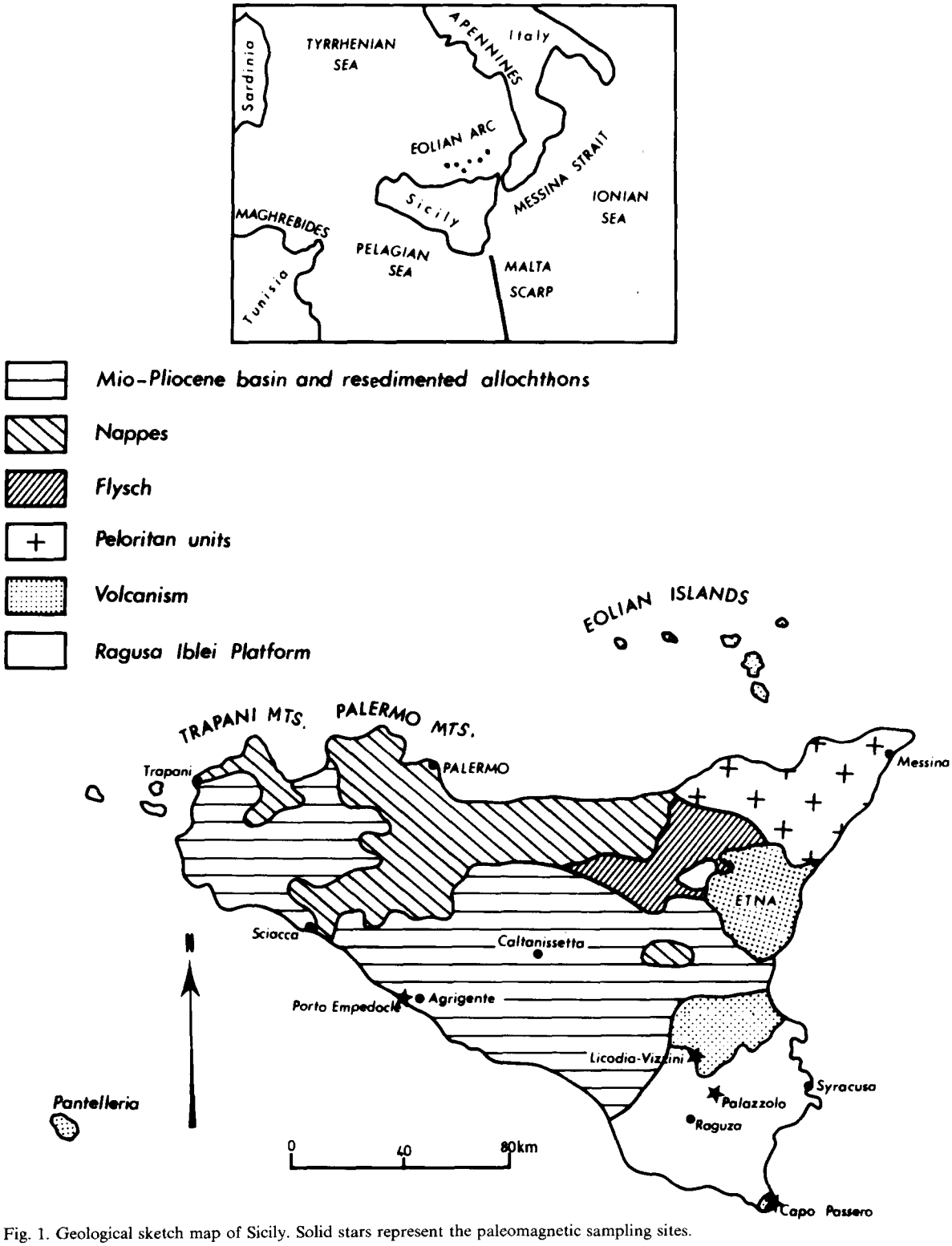


Fig. 1. Geological sketch map of Sicily. Solid stars represent the paleomagnetic sampling sites.

form systems. The age of volcanism is Pliocene and Quaternary [4]. The relation with peninsular Italy is accomplished through the Messina Strait, a recent narrow trough of N030 direction, sitting on Peloritan Calabrese units and flysch nappes, producing the Malta scarp, marked by a N170 fault, and separating Sicily from the Ionian Basin. The evidence of a volcanic arc in an advanced stage of evolution is shown by a calc-alkaline volcanic chain (Eolian Islands) and a seismic zone. Most earthquakes occur at depths ranging from 220 to 480 km [5]. This is interpreted as the sinking of a slab in the upper mantle, with a northwest dip [6]. To the north, the Tyrrhenian Sea appears to be a Neogene marginal basin. This arc differs from classic arcs by its curvature (about 100 km) and could be due to rigid plastic squashing of the African peninsula against the European plate [7].

3. Structural units of Sicily

The structural units of Sicily belong to four groups (Fig. 1):

—*Peloritan units*. The Peloritan units comprise a crystalline basement which was deformed and metamorphosed during Hercynian times and a Mesozoic/Paleogene sedimentary cover which was deformed early in the Middle Eocene. The series are unconformably overlain by the Oligocene/Early Miocene Molassic Series. This sequence is thrusted over flyschoid units and represents part of a greater unit, the Calabro-Peloritan block, thrusted over ophiolites [9], the latter being thrusted on the Apenninic units [2]. A consequence of the Tyrrhenian Sea opening is the slipping of this block toward the east-southeast [10].

—*The flyschoid units*. Several flyschoid units ranging from Lower Cretaceous to Lower Miocene have been distinguished in Sicily [10]. The units have great affinities with the North African flysch nappes.

—*The Apenninic units*. These units comprise calcareous or marly-calcareous series of Permian to Miocene age [2]. They form a series of nappes stacked during various tectonic phases.

—*The foreland*. It comprises the Iblean domain or Ragusa platform. The stratigraphic sequence ranges from Triassic to Pliocene and includes marly-calcareous sequences. Characteristic platform facies only appears during the Upper Triassic/Lower Lias and also during the Miocene, which differentiates this domain from the Apulian platform on one hand and from the Pelagian platform bordering Africa on the other hand. Stratigraphic analogies, in particular alkaline volcanics of Triassic, Middle Jurassic and Upper Cretaceous age, underline the paleogeographic similarity of this domain to the Apenninic units. This domain is not very deformed. The analysis of paleostress by microtectonic studies [11] shows two compressional phases which can be also found in Tunisia, but most structures originate from extensional faults of direction N030, 040 and N120, 150. North of this domain, a volcanic complex (Monte Lauro) was active from the Upper Miocene to Quaternary, and indicates intracontinental distension [12]. This system was replaced by Mount Etna during the Quaternary, showing a migration of the active centers toward the north.

4. Paleomagnetic data available for Sicily

Several studies have been carried out on volcanics or sedimentary series from the northern nappes [1,13]. These results have been obtained on allochthonous formations and have been used to discuss the acquisition of curvature in the Tyrrhenian arc.

The results from the Late Cretaceous and Plio-Pleistocene volcanics from the autochthonous Ragusa Iblei platform [12,14,15] will be included in this study.

5. Paleomagnetic data

Sampling localities and lithostratigraphic characteristics are summarized respectively in Fig. 1 and Table 1. Samples were collected either as hand blocks or by drilling. Orientations have been taken by magnetic compass or by sun-compass when possible. Cylindrical standard core specimens have been drilled from hand samples in the laboratory. Nannofossils and planktonic foraminifera have

TABLE 1

Micropaleontological age of principal formations

Sites	Age	Lithology	Calcareous nannofossils		Planktonic foraminifera	
Palagonia Gela	Sicilian	blue marls			N23/ N22	*Globorotalia truncatulinoides* Zone
Licodia Eubea	Zanclean	calcareous marls volcanic ashes	NN13	*Ceratolithus rugosus* Zone	PL2	*Globorotalia puncticulata* Zone
			NN12	*C. Tricorniculatus* Zone	PL1	*Globorotalia margaritae* Zone
Tellaro	Lower Tortonian	marls and calcareous marls	NN10	*Discoaster calcaris* Zone		
Vizzini	Lutetian	fine-grained semi-pelagic limestones and interlayed marls	NP15	*Chiphragmalithus alatus* Zone	P12	*Globorotalia lehneri* Zone

Biozones defined by Martini [41], Blow [42] and Berggren [43].

been used for the age determinations. Measurements have been carried out on two spinner magnetometers (Schönstedt DSM1 and Jelinek JR-4), and a cryogenic magnetometer (SCT), according to the intensity of magnetization. Magnetic properties have been studied by progressive stepwise demagnetization either by alternating field (AF) or by heating. Magnetization directions have been followed during demagnetization by orthogonal projection vector diagrams [16]. For most sites, two components could be isolated: the first one, generally soft, was along the present field and

TABLE 2

Paleomagnetic data

	Site	Age	N	n	Type of formation	D (°)	I (°)	K	α_{95} (°)	Polarity	Reference
1	Licodia	TP	1	8	vs	355	46	103	5	N	this paper
2	Licodia	TP	1	6	s	353	52	16.5	16	N	this paper
3	Licodia	TP	1	4	s	352	56	43	14	N	this paper
4	Licodia	TP	1	7	s	354	58	68	7	N	this paper
5	Licodia	TP	1	3	s	173.4	−57			R	this paper
6	Porto Empedocle	TP	2	17	s	228	−50	25	7	R	this paper
7	Tellaro	TM	1	7	s	180	−51	62	8	R	this paper
8	Tellaro	TM	1	3	s	166	−44.5	87	13	R	this paper
9	Tellaro	TM	1	7	s	166	−51	20	13	R	this paper
10	Monte Iblei	TP/TQ	10	84	v	356	48	24	10	M	Barberi et al. [12]
11	Monte Iblei	TP/TQ	19	100	v	161	−50	22	8	M	Gregor et al. [15]
12	Vizzini	TE	2	33	s	352	35.5	140	2.3	N	this paper
13	Capo Passero	KU	8	33	v	165	−26	62	7	R	Barberi et al. [12]
14	Capo Passero	KU	15	80	v	163	−30	13	9.8	R	Gregor et al. [15]
15	Capo Passero	KU	19	107	v	167	−22	23	7	R	Schult [14]

Age: TP = Pliocene, TM = Miocene, TQ = Quaternary, TE = Eocene, KU = Upper Cretaceous; N = number of sites; n = number of samples; Type of formation: v = volcanic, vs = ash, s = sedimentary; Polarity: N = normal, R = reverse, M = mixed.

easily removed, the second one, harder, generally defined a characteristic direction of magnetization. For each formation and lithology, the isothermal remanent magnetization has been measured up to saturation (IRM_s). For each step of heating, a specimen has been magnetized up to saturation and the IRM_s acquired has been analysed by AF demagnetization. Mineralogical changes of magnetic minerals are evidenced, fixing the range of the thermal demagnetization. The paleomagnetic results will be presented in chronological order, going from ancient to recent formations (see also Table 2).

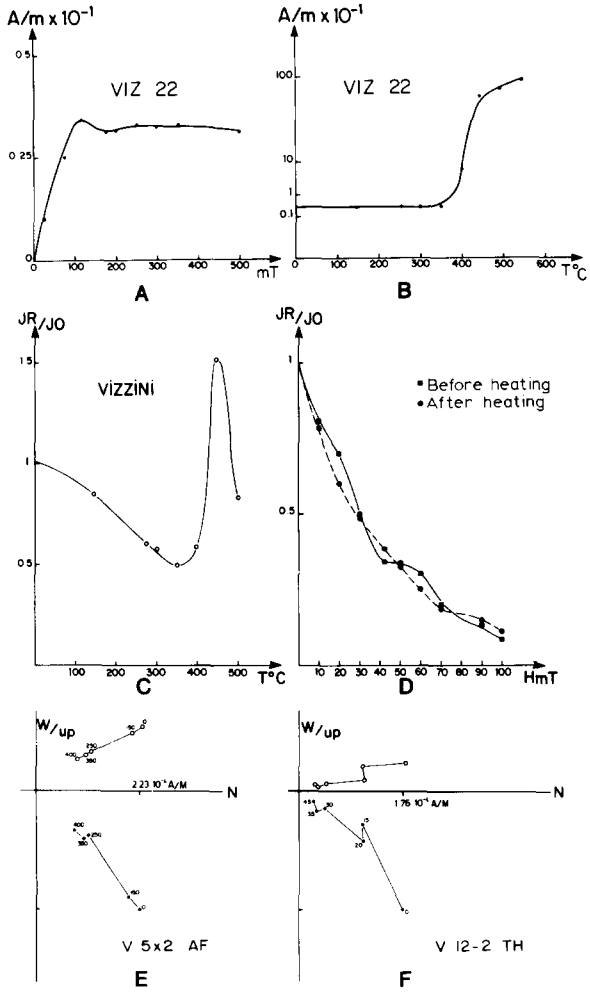


Fig. 2. A. IRM acquisition curve for Vizzini samples. B. Saturation of the isothermal remanent magnetization (IRMs) acquired and measured after each step of heating. C. Normalized thermal demagnetization curve of IRMs. D. Normalized alternating fields demagnetization curve of IRMs before and after heating at 500°C. Vector diagrams of progressive thermal (E) and alternating fields (F) demagnetization of Vizzini samples. Open symbols represent projection on the horizontal plane and stars are those on the north-south plane.

5.1. *Semipelagic limestone of Lutetian age (Vizzini)*

Thirty-three samples have been taken from two sites. The NRM intensities are close to 5×10^{-4} A/m, well grouped in a direction distinct from the present dipole field both before or after tectonic correction. The IRM_s acquisition (curve Fig. 2A) shows that saturation is reached in a field of about 200 mT indicating that magnetic minerals are probably titanomagnetites or sulfides. IRM_s measured after each temperature show a sharp increase at 350°C which probably reflects a mineralogical transformation (Fig. 2B). The thermal demagnetization of IRM_s acquired before heating clearly shows the same transformation by a sharp increase at 400°C (Fig. 2C), although AF demagnetization of IRM_s before heating and after heating are not very different (Fig. 2D). The changes of magnetic mineralogy, and particularly the sharp increase of IRM_s after 400°C, give rise to parasitic TRM and VRM acquired in a very small residual field or by interaction between grains.

The characteristic direction of magnetization obtained by heating (Fig. 2E) is the same as that obtained by AF (Fig. 2F) but with better accuracy. The characteristic direction of magnetization averaged at the sample level is: $N = 2$, $n = 33$, $D_m = 352°$, $I_m = 35.5°$, $K = 140$, $\alpha_{95} = 2.3°$.

5.2. *Tortonian marls and calcareous sediments from the Tellaro Formation*

Five sites of Upper Langhian to Lower Tortonian age have been sampled. The intensity of magnetization of two Langhian sites was very weak (5×10^{-6} A/m) and no demagnetization could be performed; these sites had to be rejected. The intensity of magnetization for the Tortonian marls was of the order of 0.5×10^{-4} A/m and measurement was performed using a SCT cryogenic magnetometer. The saturation of IRM is obtained at 250 mT (Fig. 3A). The IRM_s curve after step

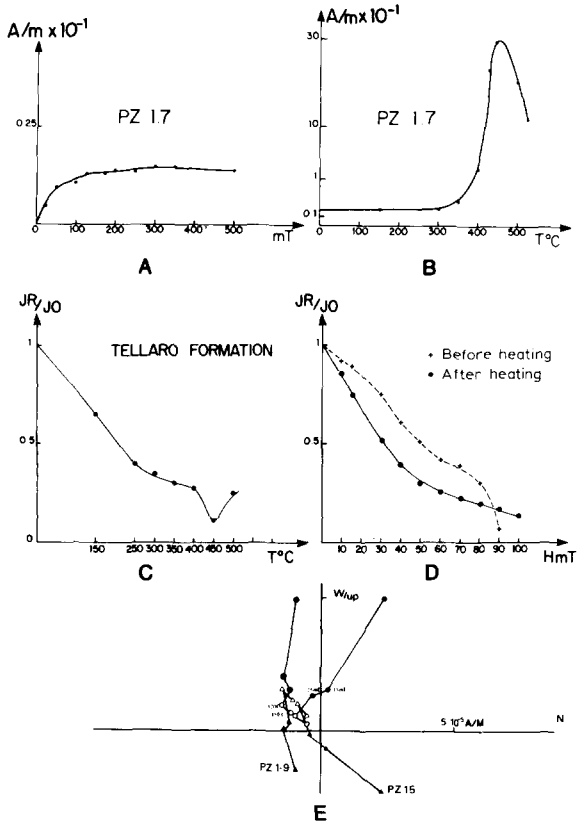


Fig. 3. Tellaro Formation. A, B, C. D, same legend as for the corresponding figures in Fig. 2. E. Composite demagnetization vector diagram for two samples. Circles represent the projection on the horizontal plane and triangles the projection on vertical plane. Solid symbols correspond to progressive alternating fields demagnetization, open symbols to thermal demagnetization.

heating (Fig. 3B) shows a sharp increase after 350°C which is also visible on thermal demagnetization of IRM_s produced before heating (Fig. 3C). The AF demagnetization of IRM_s before and after heating are quite different and shows a change in the coercivity spectrum (Fig. 3D). We have performed a two-stage demagnetization, first by AF up to 250 mT then by heating up to 200°C. A reversed component of magnetization has been clearly isolated (Fig. 3E). The characteristic direction of the magnetization is: $N = 3$, $n = 17$, $D_m = 171°$, $I_m = -49°$, $K = 137$, $\alpha_{95} = 10°$

5.3. The Pliocene Trubi Formation

The Pliocene Trubi Formation contains volcanosediments interbedded with calcareous marls. The volcanic ashes originate from Monte Iblei eruptions. Six sites have been sampled in marls and volcanosediments. Sediments have been first heated to 120°C and have shown a sharp decrease of magnetization due to the removal of soft components. The next step of heating at 150°C does not change the direction but unfortunately during the following steps the intensity of magnetization drops below the level of sensitivity of the magnetometer. The IRM is saturated for field intensities of the order of 150 mT which precludes an appreciable amount of hematite (Fig. 4A). The

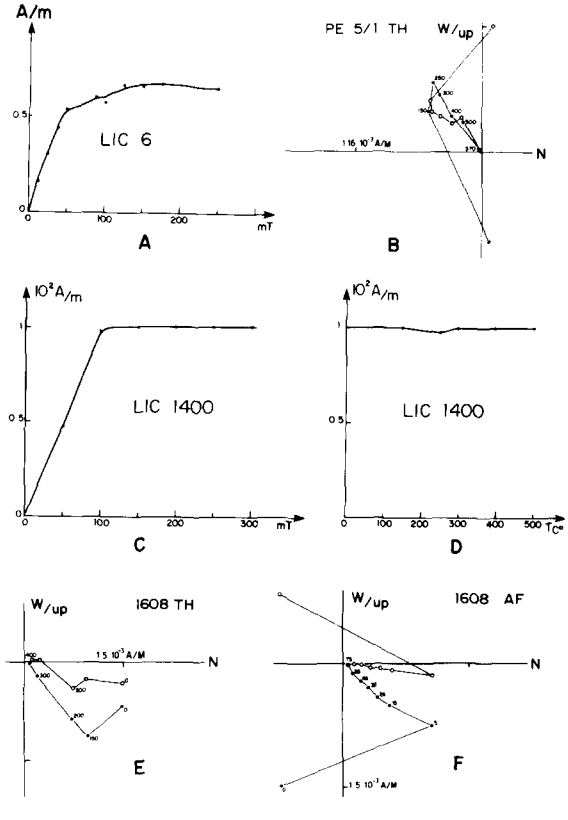


Fig. 4. IRM acquisition curve for Licodia sedimentary samples. B. Vector diagram of progressive thermal demagnetization for Porto Empedocle samples. Symbols as in Figs. 2. and 3. C. IRM acquisition curve for Licodia volcanosediments. D. Saturation of the isothermal (IRM_s) acquired and measured after each step of heating. Vector diagrams of progressive thermal (E) and AF (F) demagnetization for the same samples.

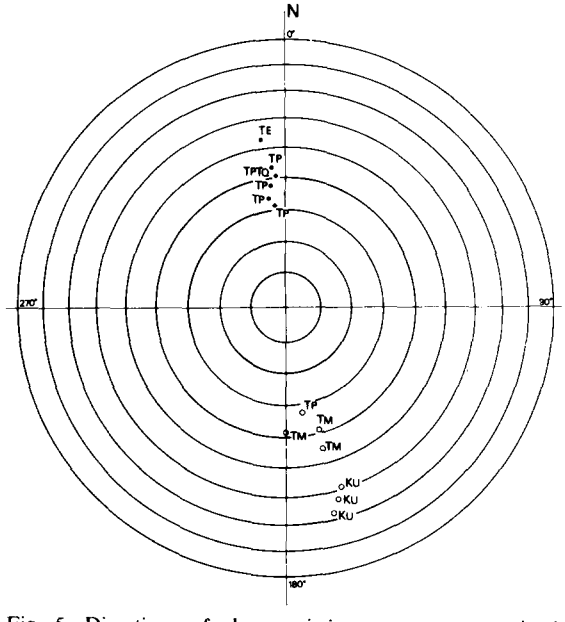


Fig. 5. Directions of characteristic remanent magnetization (ChRM) for all available sites after bedding corrections. Solid dots refer to normal polarity, open dots to reverse polarity. TQ = Pleistocene, TP = Pliocene, TM = Miocene, TO = Oligocene, TE = Eocene, KU = Upper Cretaceous.

volcanosediments have been analysed by AF and stepwise heating. Both treatments show a stable component (Fig. 4E and F) and have been demagnetized up to 75 mT and 400°C. The IRM is saturated at 150 mT (Fig. 4C) and no change in magnetic mineralogy occurs before the highest temperature of demagnetization (Fig. 4D). The site sampled in the lower part of the series, just above the Mio-Pliocene transition is of reverse polarity: this is in good agreement with the known reversal at the Mio-Pliocene transition. The directions of magnetization of reverse and normal sites are clearly opposite. Although only very slight tilts are involved (10–15°), the bedding correction improves the clustering of the directions ($N = 5$):

before tectonic correction—$D_m = 353°$,
$I_m = 64°$, $K = 105$, $\alpha_{95} = 7.5°$
after tectonic correction—$D_m = 353.5°$,
$I_m = 54°$, $K = 167$, $\alpha_{95} = 4.5°$

Two complementary sites have been sampled in the Pliocene of Porto Empedocle (Caltanissetta Basin). We have thermally demagnetized these samples up to 570°C (Fig. 4B). The mean declination ($D_m = 228°$) clearly indicates that the sampled units have been rotated relative to the Ragusa platform, but the inclinations are in close agreement thus providing a test on the validity of the computed paleolatitude during the Pliocene. The directions of magnetization for all sites are given in Table 3 and plotted, after bedding correction, on a equal-area stereonet (Fig. 5).

6. An apparent polar wander path for Sicily since the Upper Cretaceous

We have combined our results and those of Barberi et al. [12], Gregor et al. [15] and Schult [14] to compute the apparent polar wander path (APWP) presented in Fig. 6 and Table 3. Volcanic rocks from older studies have been used to compute the Plio-Pleistocene and the Upper Cretaceous pole. In order to obtain the best possible integration of secular variation, we have averaged all site directions for each age. Some of these directions were too scattered ($\alpha_{95} > 15°$) and have

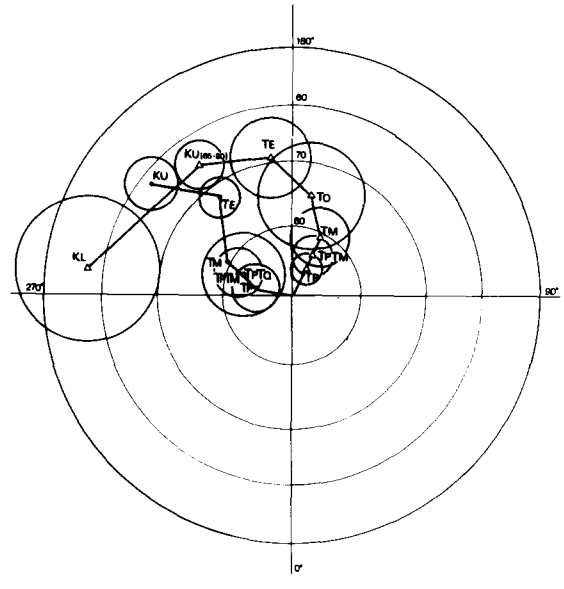


Fig. 6. Apparent polar wander path for Sicily (black dots) and Africa (triangles). Same age labels as before. Confidence ellipses have been replaced by circles with a diameter equal to the major axis of the ellipse.

TABLE 3

Upper Mesozoic and Cenozoic poles for Sicily and Africa

Age	Description	Observed directions for Sicily								Expected directions							Relative motion	
		pole position lat., long.	field direction							pole position			expected field				ΔD counter-clockwise	ΔD southward
			D (°)	I (°)	K	α_{95} (°)	λ	$\delta\lambda$	δD	lat.,	long.	A_{95}	D	δD	λ	$\delta\lambda$		
TP/TQ	mean of 28 sites from 10–11	82, 253	351.5	52	17	7	32.5	6.5	11	86,	145	1.8	4	2	35	1.5	12.5 ± 11	2.5 ± 6.5
TP	mean of sites 1–5	84, 260	353.5	54	267	4.5	34	4.4	7.5	86,	149	2.6	4	3	34	2.5	10.5 ± 8	0 ± 5
TP/TM	mean of sites 1–5, 7–9	82, 250	352.5	52	196	4	32	3.5	6.5	84,	152	3	5	3.5	33	3	12.5 ± 7	1 ± 4.5
TM	mean of sites 7–9	79, 245	171	−49	137	10	30	8.5	15	81,	154	5.6	7	6.5	30	5	16 ± 16	0 ± 10
TO										75,	170	6.5	7	7	23.5	5		
TE	mean of samples from two samples [12]	71, 218	352	35.5	140	2.5	19	1.5	3	69,	189	5.8	2	6	16.2	4.5	10 ± 6.5	−2.8 ± 4.5
KU	mean of 37 sites from 13–15	62, 231	164	−26	38.4	3.9	14	2.	4.4	65,	216	4.5	351	4.6	13.5	2.6	7 ± 6	−0.5 ± 3
KL										57,	263	12						

Age: TP = Pliocene, TQ = Quaternary, TM = Miocene, TO = Oligocene, TE = Eocene, KU, KL = Upper, Lower Cretaceous.

African Lower Cretaceous poles are from Gidskehaug et al. [44], Gough and Opdyke [45], Briden [46], Upper Cretaceous poles from Andriamirado [47], El-Shazli and Krs [48], Storetvedt et al. [21], Eocene pole from Hussain et al. [22], Oligocene pole from Schult [28], Mio-Pliocene pole from Tauxe et al. [31], Plio-Quaternary pole from Schult and Soffel [49], Reilly et al. [50], Watkins [51], Ade-Hall et al. [23] and Piper and Richardson [52].

$\delta D = \sin^{-1}(\sin A_{95}/\sin P_0)$ where P_0 is the paleocolatitude.

$\delta P = \alpha_{95}(1 + 3\cos^2 P_0)/2$

The errors in ΔP and ΔD are given by $\Delta P = (\delta\lambda^2_{obs} + \delta\lambda^2_{exp})^{1/2}$ and $\Delta D = (\delta D^2_{obs} + \delta D^2_{exp})^{1/2}$.

been rejected. We have also rejected some of the sites where the directions were obviously intermediate.

We have computed a mean Late Tertiary pole using Pliocene and Miocene directions from the sedimentary rocks. The tilt correction applied to eight sites gives a positive fold test at the 99% probably level [17] ($K_2/K_1 = 5.2$, K_2 and K_1 being respectively the precision parameters after and before tectonic correction). The Eocene pole has been computed using the mean of directions at the sample level.

7. Reappraisal of African polar wander path

In order to describe possible displacements of autochthonous Sicily, it was necessary to use Africa as a frame of reference. New paleomagnetic data have been obtained for Africa in the last few years and the age determinations of previously available data have been refined. These advances, along with a better understanding of the tectonic relations between Africa and Madagascar led us to propose a new evaluation of the APWP of the African continent for the Cenozoic time. All the directions considered have been obtained from demagnetized specimens of well-dated stable formations; moreover, we have not considered pole determinations associated with an α_{95} confidence interval larger than 10°.

This compilation differs from previous evaluations [18–20] as follows: We have included recent results not available in the G.J.R.A.S. list of poles —two Miocene poles and one Cretaceous pole from the Canary Islands, one Cretaceous pole from Egypt. We have considered new age determination of Oligocene poles from Libya [23,24] previously dated as Pliocene, and Oligo-Miocene poles from Ethiopian traps [23–27] previously correctly determined by Schult [28] but then considered as Eocene. We have included a late Cretaceous pole from Madagascar because a recent geophysical survey has shown that no appreciable movement occurred between Madagascar and Africa since the Early Cretaceous (M8 anomaly) [29,30]. Finally, poles from regions suspected to have suffered relative movements with respect to the African continent have been rejected (Danakil, Afar, Morocco). Mean poles have been computed for the periods indicated in Table 3. The paleolatitudes predicted by this APWP are in excellent agreement with those obtained from Leg 73 sediments [31]. The Sicilian and African APWP have been plotted in Fig. 6.

8. Discussion

Due to the lack of data for the African continent for the Upper Mesozoic and Cenozoic, many authors were obliged to use transferred directions as done by Van der Voo and French [32]. The re-evaluation of African APWP gives a reference to assess precisely the movements of blocks and microblocks in the Mediterranean area. In order to analyse the relative movements between Sicily and Africa we have used in Fig. 7 a paleolatitude vs. declination diagram [33], on which Sicilian declinations and paleolatitudes are compared to those predicted for Africa at $\lambda = 37°N$, and $\varphi = 19°E$). This diagram provides an immediate comparison of relative north-south movement between the two blocks. The sense of rotation is given by the changes of declination: an increase from west to east reflects a clockwise rotation of the block in consideration.

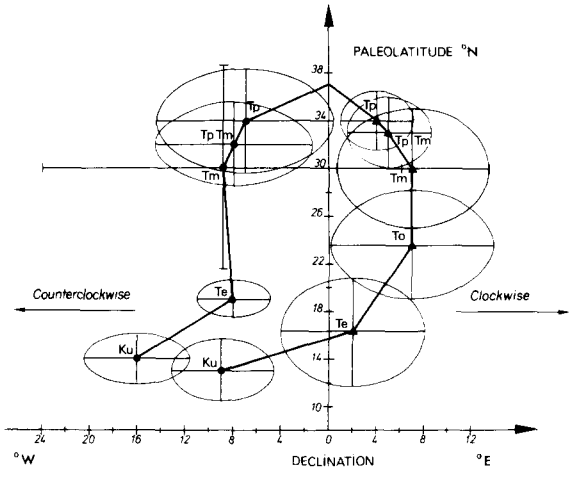


Fig. 7. Paleolatitude declination diagram for Sicily (leftmost curve) and Africa. African data have been reduced to $\lambda = 37°N$ and $\phi = 15°E$. Same age conventions as Fig. 6. The diagram was introduced in Besse [33].

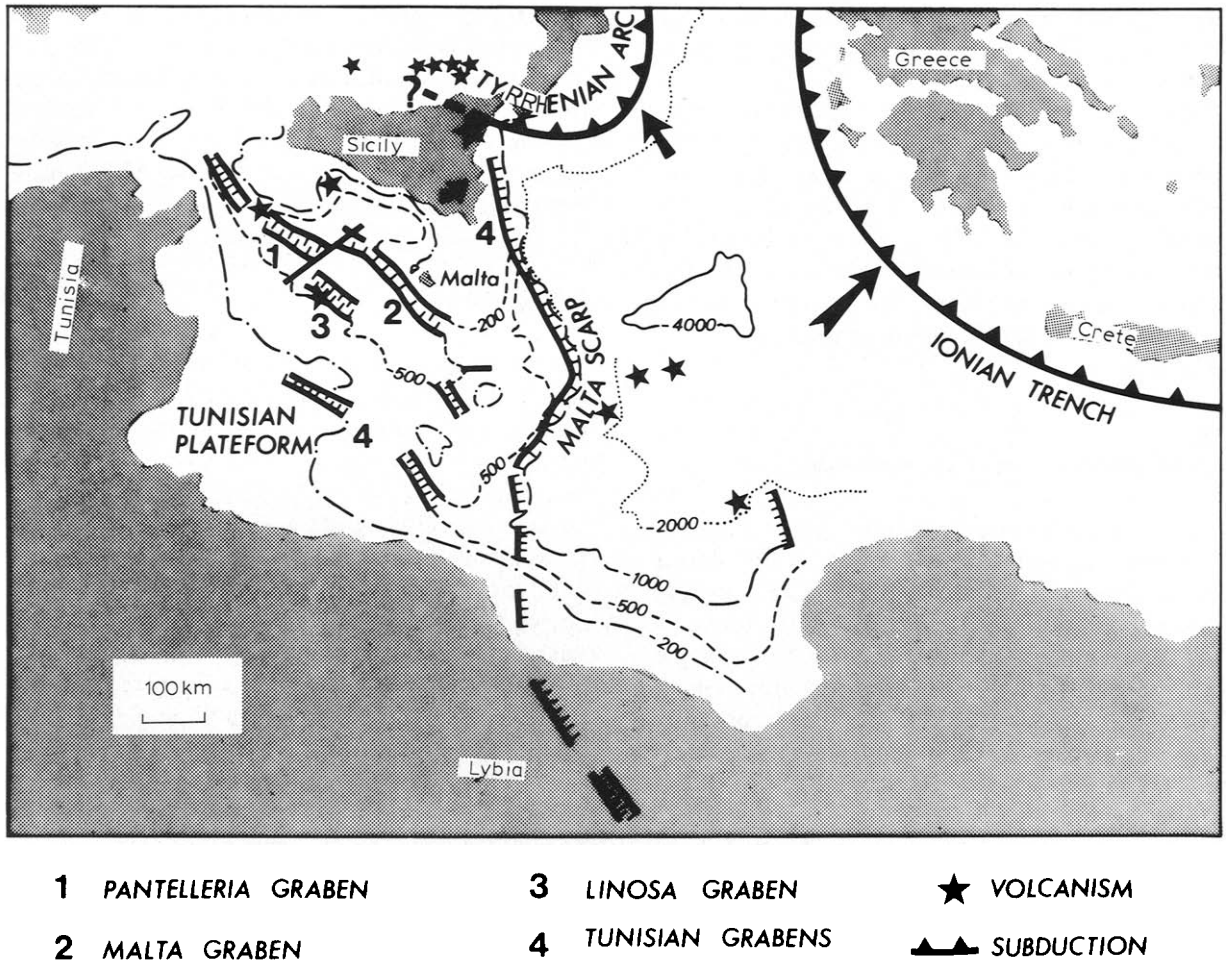


Fig. 8. Tectonics sketch map of the Pelagian Sea. The two arrows indicate the approximate directions of relative motion in the Hellenic and Tyrrhenian arcs. The end of subduction in the Tyrrhenian arc induces tensional tectonics in the Pelagian Sea marked by alcaline volcanism and the opening of "en relais" grabens widen from east to west.

During the Cretaceous, the African plate mainly undergoes an anticlockwise rotation with minor change in latitude, which can be seen as a consequence of the opening of the South Atlantic. At the beginning of the Tertiary, one can see a major northward drift, while anticlockwise rotation continues. Since the Oligo-Miocene, the sense of rotation reverses while the northward drift continues. The latitudinal evolution of the Sicilian block is identical to that of Africa since the Upper Cretaceous. But analysis of declination (Table 3 clearly shows a counterclockwise rotation relative to Africa since Late Miocene. The amplitude of this rotation now reaches 10–15°. This result is consistent with contemporaneous extensional tectonics in the Pelagian Sea, as shown by the alkaline Plio-Quaternary volcanism, steep normal faults and "en relais" grabens widening from west to east (Fig. 8). The geometry of grabens and the constraint in paleolatitude place the Sicilian/African eulerian pole of rotation near the western tip of Sicily. This rotation can be seen as a result of the mechanical disymmetry induced on one hand by the general continental blocking-up toward the west following the Miocene collision phases in Western Europe, and on the other hand by the continuation of an active subduction under the Aegean Arc. However, a possible effect of the

TABLE 4

Selected Italian data from the Upper Cretaceous to the present

Age	Description	Pole position			Reference
		lat., °N	long., °E	A_{95} (°) (dp,dm)	
Upper Cretaceous	Africa	65	216	4.5	this paper
	Sicily	62	231	2.3, 4.2	this paper
	Apulia	63	236	2.8, 5	Channell and Tarling [34]
	Gargano	61	249	4.5, 7.6	Channell [35]
Eocene	Africa	69	189	5.8	Hussain [22]
	Sicily	71	218	1.7, 2.8	this paper
	Umbria	54	260	1.5, 2.4	Lowrie et al. [39]
Oligocene	Africa	75	170	6	Schult [28]
	Umbria	65	247	2.4, 3.8	Lowrie et al. [39]

nearby dying subduction of the Calabrian Arc cannot be disregarded. This had led us to consider here the evolution of the western Italian margin since the Upper Cretaceous. Many Upper Mesozoic paleomagnetic results for the Italian Peninsula have been published during the last decade (Table 4) which raise some problems. Problems with age determinations, such as the age of diagenesis of Apulian bauxites, for which the pole cannot be used. Limestones from the Gargano Basin [35] cover a large time span (Turonian to Senonian). The paleomagnetic data from Umbria have a well-constrained age, but they come from a series which is not unquestionably autochthonous [36–38].

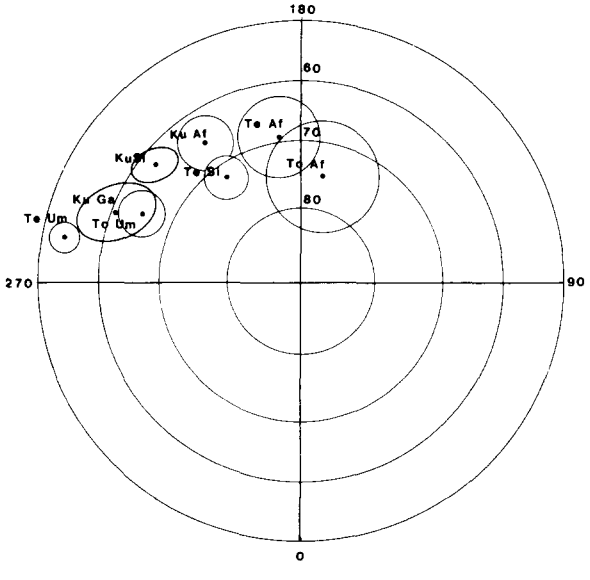


Fig. 9. A comparison between Italian, Sicilian and African poles. *Af* = Africa, *Si* = Sicily, *Um* = Umbria, *Ga* = Gargano.

The magnetostratigraphy of the Contessa section in Umbria [39] has provided two poles of Eocene and Oligocene age. These poles have also been plotted in Fig. 9 and show a major post-Oligocene phase of anticlockwise rotation. No Tertiary datum is available for the Apulian platform.

Most Italian data present common characteristics. Cretaceous inclinations indicate that peninsular Italy and Sicily were parts of Africa at least since that time. All declinations (except those of Channell et al. [1] on Sicilian northern nappes) are toward the west with, however, various amplitudes. The resulting directions are significantly different from our Upper Cretaceous African pole, and it can be therefore deduced that an anticlockwise rotation with respect to Africa occurred during the Tertiary. One of the major problems is to determine the exact time of these rotations, in order to recognize either a single rotation of the whole Italian Peninsula considered as a single crustal block, as argued by Van den Berg [36] or Channell [35], or several phases of rotation as the result of progressive intracontinental deformation due to the Italian/European collision. Moreover,

these rotations probably imply several levels of decoupling in the lithosphere. Some units are clearly nappes such as the north Sicilian or south Apennines units [1], whereas the structural position of some others, such as Umbria, is not really clear and could correspond to décollements on Triassic evaporites. It can be noticed that the rotation phases of Umbria is synchronous with the progressive suturation of the Corsica/Sardinia microplate against the Italian margin [40], from north to south, between the Oligocene and recent times. Therefore, the counterclockwise rotation of Umbria can be seen as the deformation of a surficial part of the margin during the progressive collision. On the contrary, the autochthonous Gargano Basin, and the Apulian platform may involve a thicker part of the lithosphere, as well as the Sicilian foreland where we can observe volcanism and opening of grabens in the Pelagian Sea. The Gargano pole appears to be significantly different from the Sicilian one. However, due to the large time uncertainty of the Upper Cretaceous pole of the Gargano and the lack of younger paleomagnetic data for this area, the differences in declination can be interpreted either by a relative rotation of the Sicilian and Apulian platform or by a difference of age between the two poles.

9. Conclusion

Paleomagnetism of sediments from the Iblean platform has shown that a differential rotation between autochthonous Sicily and Africa occurred since the Upper Miocene. This rotation has an amplitude of 15° which is in good agreement with contemporaneous extensional tectonics in the Pelagian Sea. The comparison with Upper Mesozoic and Cenozoic data from peninsular Italy shows that, if the role of Italy as an indenter remains clear, it seems hazardous to consider the whole peninsula as an undeformable unit. The collision most likely resulted in a strong deformation, marked both by surface rotations of tectonic units (thrusting or décollements), and by relative rotations of 100-km-sized microblocks, leading to a possibly progressive break-up of the Italian promontory.

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References

1 J.E.T. Channell, R. Catalano, and B. d'Argenio, Paleomagnetism and deformation of the Mesozoic continental margin in Sicily, Tectonophysics, 61, 391–407, 1980.
2 C. Grandjacquet and G. Mascle, The structure of the Ionian Sea, Sicily and Calabria-Luciana, in: The Ocean Basins and Margins, Vol. IVB. The Western Mediterranean, A.E.M. Nairn, W.H. Kanes and F.G. Stehli, Ed., pp. 27–29, 1978.
3 B. Colombi, P. Giese, G. Luongo, C. Morelli, M. Riuscetti, S. Scarascia, K.G. Schutte, J. Strowald and G. Vinsentini, Preliminary report on the seismic refraction profile Gargano Salerno-Palermo-Pantelleria, Boll. Geofis., 1973.
4 F. Barberi, S. Borsi, G. Ferrera and F. Innocenti, Strontium isotopic composition of some Recent basic volcanites of the Southern Tyrrhenian Sea and Sicily Channel, Contrib. Mineral. Petrol., 23(2), 1969.
5 E. Peterschmitt, Quelques données nouvelles sur les séismes profonds de la mer Tyrrhénienne, Ann. Geophys. 9, 305–334, 1956.
6 A.R. Ritsema, Active or passive subduction at the Calabrian arc, Géol. Mijnbouw 58(2), 127–134, 1979.
7 P. Tapponnier, Evolution tectonique du système alpin en Méditerranée: poinçonnement et écrasement rigide plastique, Bull. Soc. Géol. Fr. (7), XIX(3), 437–460, 1977.
8 A. Caire and R. Truillet, A propos de la phase tectonique tertiaire antérieure au dépôt de l'Oligo-Miocène dans les monts Péloritains (Sicile), C.R. Acad. Sci. 256, 2446–2447, 1963.
9 R. Dubois, La structure calabro-apenninique Crétacé-Eocène et l'ouverture tyrrhénienne néogène: étude pétrographique et structurale de la Calabre centrale, 567 pp., Thèse Sciences, Paris, 1976.
10 P. Broquet, G. Duee, G. Mascle, and R. Truillet, Evolution structurale récente de la Sicile: signification géodynamique, Rev. Géol. Dyn. Phys. (in press).
11 J. Letouzey, and P. Tremolières, Paleostress field around the Mediterranean since the Mesozoic derived from microtectonics: comparison with plate-tectonic data, in: Les chaînes alpines issues de la Téthys, J. Aubouin, J. Debelmas and M. Latreille eds., Collogr. C6, 26th Int. Geol. Congr., Mém. B.R.G.M., 115, 261–273, 1980.
12 F. Barberi, L. Civetta, P. Gasparini, F. Innocenti, R. Scandone, and L. Villari, Evolution of a section of the Africa-

Europe plate boundary: paleomagnetic and volcanological evidence from Sicily, Earth Planet. Sci. Lett. 22, 123–132, 1974.
13 A. Schult, Paleomagnetism of Cretaceous and Jurassic volcanic rocks in Western Sicily, Earth Planet. Sci. Lett. 31, 454–457, 1976.
14 A. Schult, Paleomagnetism of Upper Cretaceous volcanic rocks in Sicily, Earth Planet. Sci. Lett. 19, 97–100, 1973.
15 G.B. Gregor, A.E.M. Nairn and J.F.W. Negedank, Paleomagnetic investigations of the Tertiary and Quaternary rocks IX. The Pliocene of southeast Sicily and some Cretaceous rocks from Capo Passero, Geol. Rundsch. 64(3), 948–957, 1975.
16 Zijderveld, J.D. Demagnetization of rocks: analysis of results, in: Methods in Paleomagnetism, D.W. Collinson, K.M. Creer and S.K. Runcorn, eds., Elsevier, Amsterdam, 1967.
17 M.W. McElhinny, Statistical significance of the fold test in paleomagnetism, Geophys. J.R. Astron. Soc. 8, 338–340, 1964.
18 M. Westphal, Contribution du paléomagnétisme à l'étude des déplacements continentaux autour de la Méditerranée Occidentale, Thèse Université Louis Pasteur, Strasbourg, 1973.
19 P.K.S. Raja, Paleomagnetism of the Cenozoic volcanic rocks from east Africa, Curr. Sci. 45(2), 46–48, 1976.
20 J. Van den Berg, Paleomagnetic data from the western Mediterranee: a review, Geol. Mijnbouw 58(2), 161–174, 1979.
21 K.M. Storetvedt, V. Mounstad, S. Asse and R. Lovlie, Paleomagnetism and the early magmatic history of Fuerteventura (Canary Islands), J. Geophys. 46, 319–331, 1979.
22 A.G. Hussain, A. Schult and H. Soffel, Paleomagnetism of basalts of Wadi Abu Tereifiya, Mandisha and Dioritic dikes of Wadi Abu Shiat, Egypt, Geophys. J. R. Astron. Soc. 56, 55–61, 1979.
23 J.M. Ade-Hall, S. Gerstein, R.E. Gerstein, P.H. Reynolds, P. Dagley, A.E. Mussett and T.P. Hubbard, Geophysical studies of North African Cenozoic volcanic areas, III. Garian, Libya, Can. J. Earth Sci. 12, 1264–1271, 1975.
24 J.M. Ade-Hall, P.H. Reynolds, P. Dagley, A.E. Mussett and T.P. Hubbard, Geophysical studies of North African Cenozoic volcanic areas, II. Jebel Soda, Libya, Can. J. Earth Sci., 12, 1257–1263, 1975.
25 D.C. Rex, I.L. Gibson and F. Dakin, Age of the Ethiopian flood basalt succession, Nat., Phys. Sci. 230, 131–132, 1971.
26 P.W. Jones and D.C. Rex, New dates from the Ethiopian plateau volcanics, Nature 252, 218–219, 1974.
27 I. McDougall, W.H. Morton, and M.A.J. Williams, Age and rates of denudation of trap Series basalts at Blue Nile Gorge, Ethiopia, Nature 254, 207–209, 1975.
28 A. Schult, Paleomagnetism of Tertiary volcanic rocks from the Ethiopian southern plateau and the Danakil block, J. Geophys. 40, 203–312, 1974.
29 J. Segouffin, Morphologie et structure du canal de Mozambique, Thèse, Université Louis Pasteur, Strasbourg, 1980.
30 P.D. Rabinovitz and M.F. Coffin, The Madagascar-Africa separation, EOS 62(17), 297, 1981.
31 L. Tauxe, J. Besse and J. LaBrecque, Paleolatitudes from Leg 73 sediments core: implications for the apparent polar wander path for Africa during the late Mesozoic and Cenozoic, Geophys. J. R. Astron. Soc. 73, 315–324, 1983.
32 R. Van der Voo and French, Apparent polar wandering for the atlantic-bordering continents: late carboniferous to eocene, Earth-Sci. Rev. 10, 99–119, 1974.
33 J. Besse, Paléomagnétisme de séries autochtones de Sicile, 80 pp., Thèse, Paris, 1981.
34 J. Channell and R. Tarling, Paleomagnetism and the rotation of Italy, Earth Planet. Sci. Lett. 25, 117, 1975.
35 J. Channell, Paleomagnetism of limestones from the Gargano Peninsula (Italy) and the implication of these data, Geophys. J. R. Astron. Soc. 51, 605–616, 1977.
36 J. Van den Berg and A.A.H. Wonders, Paleomagnetic evidence of large fault displacement around the Po basin, Tectonophysics, 33, 301–320, 1976.
37 R. Kliegfield and J.E.T. Channell, Paleomagnetic evidence of large displacement around the Po basin—discussion, Tectonophysics 53, 139–146, 1979.
38 J. Van den Berg and A.A.H. Wonders, Paleomagnetic evidence of large fault displacement around the Po basin—reply, Tectonophysics 53, 142–146, 1979.
39 W. Lowrie, W. Alvarez, G. Napoleone, K. Perch-Nielsen, I. Premoli-Silva and M. Toumarkine, Paleogene magnetic stratigraphy in Umbrian pelagic carbonate rocks: the Contessa sections, Gubbio, Geol. Soc. Am. Bull. 33, 414–432, 1982.
40 R. Montigny, J.B. Edel and R. Thuizat, Oligomiocene rotation of Sardinia, K/Ar ages and paleomagnetic data of Tertiary volcanics, Earth Planet. Sci. Lett. 54, 261–271, 1981.
41 E. Martini, Standard Tertiary and Quaternary calcareous nannoplankton zonation, in: 2nd Planktonic Conf. Proc., 2, Proc. A. Farinacci, ed., pp. 739–785, Edizioni Tecnoscienza, Rome, 1971.
42 W.H. Blow, Late Middle Eocene to Recent planktonic foraminiferal biostratigraphy, Proc. Int. Conf. Planktonic Microfossils, Geneva 1967, 1969.
43 W.A. Berggren, The Pliocene time-scale: calibration of planktonic foraminiferal and calcareous nannoplankton zones, Nature, 243, 391–397, 1973.
44 A. Gidskehaug, K.M. Creer and J.G. Mitchell, Paleomagnetism and K-Ar ages of the South-West African basalts and their bearing on the time of initial rifting of the South Atlantic Ocean, Geophys. J. R. Astron. Soc. 42, 1–20, 1975.
45 D.I. Gough and N.D. Opdyke, The paleomagnetism of the Lupata Alkaline volcanics, Geophys. J. R. Astron. Soc. 7, 457–468, 1963.
46 J.C. Briden, Paleomagnetism of the Ntonga ring structure, Malawi, J. Geophys. Res. 73, 725–733, 1968.
47 C.R.A. Andriamirado, Thèse, University of Strasbourg, 1971.

48 E.M. El-Shazli and M. Krs, Paleogeography and paleomagnetism of the Nubian sandstones, Eastern desert of Egypt, Geol. Kunm. 62, 212–225, 1973.
49 A. Schult and H. Soffel, Paleomagnetism of Tertiary basalts from Libya, Geophys. J. R. Astron. Soc. 32, 373–380, 1973.
50 T.A. Reilly, P.K.S. Raja, E.A. Mussett and A. Brock, The paleomagnetism of Late Cenozoic volcanic rocks from Kenya and Tanzania, Geophys. J. R. Astron. Soc. 45, 483–494, 1976.
51 N.D. Watkins, Paleomagnetism of the Canary Islands and Madeira, Geophys. J. R. Astron Soc. 32, 249–267, 1973.
52 J.D.A. Piper and A. Richardson, The paleomagnetism of the Gulf of Guinea volcanic province, West Africa, Geophys. J. R. Astron. Soc. 29, 147–171, 1972.